Mohammad Amin Sadat

Embedded Software Engineer

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Professional Summary

Embedded Software Engineer with 3+ years of experience in safety-critical automotive ADAS systems (ISO 26262, ASIL B/D), holding an M.Sc. in Mechatronics Engineering. Specialises in AEB, ACC, FCW, LDW, and TSR software for BYD, JAC, and GM platforms - covering the complete V-cycle from MIL through MiL/SiL (VTD), HiL (CAN simulation), and vehicle validation with MF4 data recording. Proficient in model-based design (MATLAB/Simulink), AUTOSAR architecture, hierarchical state machines in C, and low-level driver implementation on Infineon AURIX (Tricore) with Hightec and Tasking compilers. Additional expertise in ROS-based robotics, FreeRTOS real-time systems, and applied AI/ML - including MPC+Reinforcement Learning for autonomous path tracking (M.Sc. thesis). Effective both as an individual contributor and as an on-site technical lead.

Skills

Languages: C (MISRA-aware), C++, Python, Rust, MATLAB/Simulink
Microcontrollers & Compilers: ARM Cortex-M, Infineon AURIX TC3xx (Hightec, Tasking), STM32, Embedded Linux, Arduino
RTOS & Middleware: FreeRTOS, AUTOSAR (BSW/RTE), ROS/ROS2, Linux Kernel Module
Communication Protocols: CAN/CAN-FD, LIN, FlexRay, UDS, USB, UART, SPI, I2C, LoRa, Zigbee, Wi-Fi, BLE, TCP/IP stack
Testing & Tooling: GTest (MiL), VTD (SiL), CAN-signal HiL simulation, MF4 vehicle data recording, Vector CANoe, Vector CANape, TOUSAN TSmaster, ZLG, Lauterbach, JTAG/SWD, GDB, Git, Jira, CMake
Standards & Methodologies: ISO 26262 (ASIL B/D), SOTIF awareness, AEB, FCW, LDW, TSR, ACC, Agile/Scrum
AI / ML: Deep Learning (PyTorch/TensorFlow), MPC, Reinforcement Learning, edge inference, ROS perception pipelines

Experience

Software Motion - Embedded ADAS Engineer (Decision, Planning & Control Team) — Apr 2025 - Present

- Developed safety-critical embedded software for AEB, FCW, LDW, TSR, and ACC functions on BYD, JAC, and GM vehicles; implemented hierarchical state machines in C with full AUTOSAR RTE integration.
- **Acted as de-facto technical lead for embedded C development:** the team's software lead specialised in model-based design (Simulink); assumed ownership of C coding standards, architecture decisions, and code review for state machine modules across the team - mentored colleagues on hierarchical state machine design, AUTOSAR RTE integration patterns, and embedded coding practices.
- Designed and configured CAN communication stack (DBC signal extraction); validated end-to-end signal flow and timing with Vector CANoe; resolved field bus issues across SiL and vehicle test phases.
- Executed **Module-in-Loop (MiL)** testing with GTest framework to verify individual software unit logic and edge-case behaviour before integration.
- Performed **Software-in-Loop (SiL)** testing using VTD (Virtual Test Drive) - replayed scenario videos in front of the ADAS sensor model and simulated CAN signals to validate algorithm responses in a closed-loop virtual environment.
- Conducted **Hardware-in-Loop (HiL)** validation by injecting simulated CAN signals directly into the physical ECU (AURIX TC379, Tasking compiler), verifying real-time behaviour under representative bus conditions.
- Led **vehicle test campaigns**: recorded ECU output data in MF4 format and fed logs into the company's custom SiL simulator for regression and corner-case analysis; closed feedback loop with firmware team to resolve field issues.
- **From Nov 2025 - Onsite Technical Lead:** managed multi-ECU vehicle test sessions, coordinated data logging strategy, and reduced field-to-fix cycle time through structured issue triage with the firmware team.

Avid Mechanic Hafez Sanaat - Embedded Mechatronics Engineer — Oct 2021 - Mar 2023

- Developed FreeRTOS firmware in C for a hospital bed controller and a 6-DOF robotic manipulator; implemented motor control, sensor fusion, and fault handling.
- Designed multi-layer PCBs in Altium Designer using STM32 MCUs with I2C/SPI peripheral trees; took boards from schematic to bring-up and production validation.

Teaching Assistant - Embedded & Control Systems — *Shahid Beheshti Univ. & Shiraz Univ.* — 2018 - 2024

- Led lab sessions on embedded C, Arduino, and real-time control; designed assignments on state machines and PID; mentored 30+ students per semester on hardware debugging.

Projects

Fuzzy Logic Library (Rust) - Embedded-friendly type-1 fuzzy inference engine implementing Mamdani and Sugeno methods; zero-copy design targeting resource-constrained MCUs. Published open-source. — fuzzy-logic_rs

MPC+RL Path Tracking for Autonomous Vehicles - Designed and validated a hybrid Model Predictive Control / deep Reinforcement Learning controller for autonomous lane tracking; achieved stable path following under model uncertainty and dynamic obstacle scenarios in simulation. — MSc Thesis, 2025

Inverted Pendulum on a Cart - Real-time embedded control on Arduino using PID and full-state feedback. Tuned controllers via serial debugging; achieved 2 degree steady-state error in hardware validation. — Spring 2023

IoT Garden - Full Embedded Product - Custom PCB (FreeCAD), ATmega328P firmware (C++), Wi-Fi telemetry upload, and real-time web dashboard for soil moisture monitoring - complete product lifecycle from schematic to deployment. — Fall 2021

Pneumatic Sequence Control - Programmed Siemens S7-1200 PLC in TIA Portal for a 3-cylinder sequencing system using ladder logic and HMI simulation. — Spring 2023

Education

MS — **Shahid Beheshti University**, Mechatronics Engineering — Sep 2022 - Feb 2025

- **Rank 1** in Mechatronics Engineering
- Thesis: *Path tracking control of a self-driving car using MPC and Reinforcement Learning* - combined classical and learning-based planners for robust lane-keeping under dynamic constraints
- Full tuition waiver scholarship; Top 0.5% in Iran's national MSc Mechanical Engineering entrance exam (2023)

BS — **Shiraz University**, Mechanical Engineering — Sep 2017 - Feb 2022

- Full tuition waiver scholarship throughout the programme

Certifications & Awards

- Motion Planning for Self-Driving Cars (Coursera)
- State Estimation and Localization for Self-Driving Cars (Coursera)
- Generative AI for Everyone (Coursera)
- Computer Networking (Coursera)
- Top 0.5% in Iran's national MSc Mechanical Engineering entrance exam (2023)
- Rank 1 GPA in Mechatronics Engineering at Shahid Beheshti University

Volunteering

Core Member of Robotics Club - Designed and delivered Arduino and MATLAB programming courses; introduced 50+ students to embedded systems and control fundamentals. — Shiraz University, Sep 2019 - Sep 2020